2022-2023 (SPRING) YTU CIVIL ENG. MECHANICS DIV.

STRENGTH of MATERIALS II

MIDTERM EXAM *26.04.2023*

Name:
Group:
No:
q=
M=
Sign:

Explanation: Parameters q, M
Number a: is the last digit of your student number.
Number b: is the second last digit of your student number.

Example for **q=(2+a) kN/m** Student number: xxxxxx94, a=4, b=9 then **q=6 kN/m**
Example for **M=(10+b) kNm** Student number: xxxxxx94, a=4, b=9 then **M=19 kNm**

Duration: **30 minutes** (solution + uploading) **Good luck**

3) (33p) For the beam given in the Figure,

a- Find the equation of the elastic curve by integrating the moment $\left(-EIv'' = M_x(z)\right)$.

b- Find the vertical displacement of the point B.

c- Find the rotation of the point B.

Note: The bending rigidity EI is constant.

